

Homework (Jalapeno)

- Q1.** The ancient Egyptians used $\mathbb{N}$ to represent which set of numbers?
(A) Whole numbers
(B) Integers
(C) Natural numbers
(D) Rational numbers
(E) Irrational numbers
- Q2.** The number 5 is classified as which type of number?
(A) Even and whole
(B) Odd and natural
(C) Even and integer
(D) Odd and integer
(E) Even and rational
- Q3.** The ancient Babylonians used a sexagesimal system that included which of the following numbers?
(A) Only whole numbers
(B) Only integers
(C) Only natural numbers
(D) Whole numbers and fractions
(E) Integers and fractions
- Q4.** The pattern 2, 5, 8, 11, 14, ... is an example of which type of sequence?
(A) Arithmetic sequence
(B) Geometric sequence
(C) Fibonacci sequence
(D) Harmonic sequence
- (E) Alternating sequence
- Q5.** The result of $(-3) + 5$ is which of the following?
(A) -8
(B) -2
(C) 2
(D) 8
(E) 12
- Q6.** The ancient Greeks used $\mathbb{Z}$ to represent which set of numbers?
(A) Whole numbers
(B) Integers
(C) Natural numbers
(D) Rational numbers
(E) Irrational numbers
- Q7.** The number -2 is classified as which type of number?
(A) Even and whole
(B) Odd and natural
(C) Even and integer
(D) Odd and integer
(E) Even and rational
- Q8.** The result of $(-2) \cdot (-5)$ is which of the following?
(A) -10
(B) -4
(C) 10
(D) 4
(E) 20

Open Ended Questions (FRQ)

- Q9.** The ancient Mayans used a vigesimal system to represent numbers. If a Mayan numeral represents the number x , and x is a natural number, show that x can be represented as $x = a_0 + a_1 \cdot 20 + a_2 \cdot 20^2 + \dots$ where a_i are digits in the Mayan system.
- Q10.** A historian discovers an ancient scroll with a pattern of numbers 1, 4, 7, 10, 13, 16, If the pattern continues, what is the next number in the sequence? Show your work and explain your reasoning.

Chef's Tips

- Tip 1: Allow students to use calculators for checkwork only.
- Tip 2: Emphasize the importance of showing work for free-response questions.
- Tip 3: Encourage students to use historical context to help solve problems.